

Homework for Chap 12

1. Calculate the plasmon energy for n-type GaAs with carrier concentrations of 10^{10} , 10^{15} and 10^{20} cm^{-3} .
2. Calculate the density of electrons at which $\hbar\omega_{\text{PL}} = \hbar\Omega_{\text{LO}}$ for ZnO and InAs, here $\hbar\omega_{\text{PL}}$ and $\hbar\Omega_{\text{LO}}$ are the plasmon energy and longitudinal optical phonon energy, respectively. Up to which temperatures are the electron gases degenerate?